

Orbit-Barycenter Fixed Points for Affine Rotative Maps

A substantial partial result for arXiv:1405.7872

Automated mathematics run `fa_banach_001`, lane 8

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Status. Candidate substantial partial result, likely valid, pending human review. The unrestricted nonlinear threshold questions remain open.

1 Source questions and scope

For a Banach space X , $n \geq 2$, and $a \in [0, n)$, Pongsriam and Termwuttipong define $\gamma(X, n, a)$ as the infimum of Lipschitz bounds for fixed-point-free (n, a) -rotative maps on closed convex subsets of X . Their Questions 3–6 ask for finiteness, exact values, the case $X = C[0, 1]$ with $n = 2$ and $a < 1$, and continuity in a [1, p. 4].

Other questions concerning the function γ are the following.

Q3: For a Banach space X , what is a good estimation for $\gamma(X, n, 0)$?
Is $\gamma(X, n, 0) < \infty$?

Q4: From the list (iii) given above, we know that $\gamma(C[0, 1], 2, a) \leq \frac{1}{a-1}$ for $a \in (1, 2)$. But nothing is known for $a \in [0, 1)$.
Is $\gamma(C[0, 1], 2, a) < \infty$ for some $a \in [0, 1)$?
Is $\gamma(C[0, 1], 2, a) = \infty$ for some $a \in [0, 1)$?

Q5: Can we find a precise value of $\gamma(X, n, a)$ for some X , n , and a ?

Q6: For a Banach space X and $n \geq 2$, is $\gamma(X, n, \cdot) : [0, n) \rightarrow (1, \infty]$ continuous?

Figure 1: Questions 3–6 in [1, p. 4].

This note completely classifies the affine subclass for all Banach spaces and all parameters: its threshold is always infinity. It also gives a quantified nonlinear extension. These results do not determine the unrestricted function γ .

2 Definitions and proof intuition

Let C be a nonempty closed convex subset of a Banach space. A map $T : C \rightarrow C$ is (n, a) -rotative if

$$\|T^n x - x\| \leq a \|Tx - x\| \quad (x \in C), \quad n \geq 2, \quad 0 \leq a < n.$$

Write $d_T(x) = \|Tx - x\|$ and form the length- n orbit barycenter

$$B_n x = \frac{1}{n} \sum_{j=0}^{n-1} T^j x \in C.$$

Proof intuition. For an affine map, applying T to this barycenter shifts every orbit term forward by one. All intermediate terms cancel when $TB_nx - B_nx$ is formed, leaving exactly $(T^n x - x)/n$. Rotativeness therefore reduces the displacement by the strict factor a/n . The move from x to B_nx is bounded by a fixed multiple of the current displacement, using only the Lipschitz constant. Repeating the barycenter operation gives geometrically decreasing displacement and summable moves, hence a Cauchy sequence whose limit is fixed.

The same mechanism survives for a nonlinear map if its failure to commute with its own orbit barycenter is quantitatively smaller than the remaining contraction margin.

3 Main results

Theorem 1 (Affine rotative fixed-point theorem). *Let C be a nonempty closed convex subset of a Banach space and let $T : C \rightarrow C$ be affine and k -Lipschitz for some finite $k \geq 0$. If T is (n, a) -rotative for $n \geq 2$ and $0 \leq a < n$, then T has a fixed point.*

More precisely, for every $x_0 \in C$, the iteration

$$x_{m+1} = B_n x_m = \frac{1}{n} \sum_{j=0}^{n-1} T^j x_m$$

converges to a point of $\text{Fix } T$.

Proof. Affineness and convexity give

$$TB_n x = \frac{1}{n} \sum_{j=0}^{n-1} T^{j+1} x, \quad TB_n x - B_n x = \frac{T^n x - x}{n}.$$

Consequently, with $q = a/n < 1$,

$$d_T(B_n x) \leq q d_T(x). \tag{1}$$

Since T^r is k^r -Lipschitz,

$$\|T^{r+1} x - T^r x\| \leq k^r d_T(x).$$

Thus

$$\begin{aligned} \|B_n x - x\| &\leq \frac{1}{n} \sum_{j=1}^{n-1} \|T^j x - x\| \\ &\leq \frac{1}{n} \sum_{j=1}^{n-1} \sum_{r=0}^{j-1} k^r d_T(x). \end{aligned}$$

$\underbrace{\hspace{10em}}_{M_{n,k}}$

Applying (1) recursively yields

$$d_T(x_m) \leq q^m d_T(x_0), \quad \|x_{m+1} - x_m\| \leq M_{n,k} q^m d_T(x_0).$$

The second estimate is summable, so (x_m) is Cauchy. Closedness of C gives a limit $p \in C$. As T is Lipschitz and $d_T(x_m) \rightarrow 0$, we obtain $Tp = p$. $\square$

Define the affine-restricted analogue of the source invariant by

$$\gamma_{\text{aff}}(X, n, a) = \inf \{k : \text{some affine } T \in \Phi(C, n, a, k) \text{ on a closed convex } C \subseteq X \\ \text{has } \text{Fix } T = \emptyset\},$$

with $\inf \emptyset = \infty$.

Corollary 2 (Complete affine classification). *For every Banach space X , every $n \geq 2$, and every $a \in [0, n)$,*

$$\gamma_{\text{aff}}(X, n, a) = \infty.$$

Hence every finite witness for $\gamma(X, n, a)$ must be genuinely nonlinear.

Theorem 3 (Robust orbit-barycentric criterion). *Let $T : C \rightarrow C$ be k -Lipschitz and (n, a) -rotative. Suppose that some $\eta \geq 0$ satisfies, for every $x \in C$,*

$$\left\| T(B_n x) - \frac{1}{n} \sum_{j=1}^n T^j x \right\| \leq \eta d_T(x). \quad (2)$$

If $\eta + a/n < 1$, then T has a fixed point, and the iteration $x_{m+1} = B_n x_m$ converges to one.

Proof. Combining (2) with the telescoping identity for the formal shifted barycenter gives

$$d_T(B_n x) \leq \left(\eta + \frac{a}{n} \right) d_T(x).$$

The bound $\|B_n x - x\| \leq M_{n,k} d_T(x)$ did not use affineness. With $q = \eta + a/n < 1$, the same geometric-decay and summable-step proof as in Theorem 1 applies. $\square$

4 Scope, failed full-upgrade routes, and verification

The result eliminates every affine construction from Questions 3–6 and gives a stable nonlinear neighborhood measured by (2). It does not compute the unrestricted $\gamma(X, n, a)$. In particular, at $a = 0$ the source problem already contains the fixed-point theory of Lipschitz periodic maps; a 2017 paper still asks whether a fixed-point-free 2-Lipschitz involution exists on a closed convex Banach subset [2, Question 4.2].

Eight focused routes were audited: source-signal separation; arbitrary affine operators; the robust defect extension; positive unilateral-shift constructions; an alternating affine shift on c_0 ; direct contraction of orbit increments; periodic/involution constructions at $a = 0$; and deformation arguments for continuity of γ . The two barycentric routes succeeded. Both shift families attain the forbidden ratio 2, direct increment estimates lose the contraction, and no fixed-point-free nonlinear involution or parameter deformation was obtained.

The proof is symbolic. Verification reduces to the affine telescoping identity, the finite constant $M_{n,k}$, and the geometric-series Cauchy estimate. The run’s four cheap indexes and a bounded web search through 17 August 2026 used the exact arXiv id and the phrases “affine rotative mapping”, “orbit barycenter”, and “ $\gamma(X, n, a)$ ”. The source, the 1999 rotative results, affine fixed-point searches, and the 2017 involution discussion were checked. No exact statement of Theorems 1 or 3 was found. This is bounded evidence, not a priority claim.

Human-review recommendation. Promote for expert review as a substantial partial result. Check the normalization in (2), the passage from displacement decay to a Cauchy sequence, and the novelty of the affine classification under older “mean iteration” terminology.

References

- [1] T. Pongsriam and I. Termwuttipong, *Fixed points of continuous rotative mappings on the real line*, arXiv:1405.7872 (2014).
- [2] J. Górnicki, *Fixed point theorems for Kannan type mappings*, J. Fixed Point Theory Appl. 19 (2017), 2145–2152.